

D5-V10: Precision Agriculture and Targeted Intervention

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Objective. Modeling of a sensor-equipped agricultural robot that inspects heterogeneous plots, diagnoses local crop conditions, chooses targeted interventions, avoids unnecessary treatment, and reasons over water, fertilizer, pesticide, and energy.

1. Modeling Assumptions

The field is modeled as finite plots connected by way-points. One mobile sensor-equipped robot is used. Sensing and actions are deterministic. Resources are abstract numeric units rather than the standard.

The robot follows the pipeline: [travel → sense → diagnose → treat → verify → report]

Q1 assumptions. Crop conditions are symbolic and remain static during the classical PDDL plan. The robot handles one active plot case at a time. A plot must be observed and diagnosed before treatment. Reporting closes the case. Refill/recharge are discrete service actions, to demonstrate suspend-service-return-resume process.

Q2 assumptions. Crop variables use simplified linear continuous rates. Events convert numeric thresholds into symbolic states such as deficiency, infestation, damage, or failure. Treatments are dynamic processes. Failure/damage are irreversible within one plan. Full refill loops were not locked into Q2 because combining service recovery with continuous crop processes caused ENHSP search explosion. Focus is on crop evolution, timing, threshold events, and resource decisions.

2. Q1: Basic PDDL

Types: robot, plot, waypoint. Conditions: moisture-low, pest-present, nutrient-low, growth-delayed. Local assessment facts: observed/classified predicates plus needs-water, needs-pesticide, needs-fertilizer, needs-growth-support and corresponding not-needed diagnoses. Numeric fluents represent carried water, fertilizer, pesticide, growth-support material, and battery.

Implementation pipeline.

- **Navigate/inspect:** move through row adjacency and open a local assessment case.
- **Observe:** colimitedllect moisture, pest, nutrient, and growth evidence.
- **Diagnose:** classify each dimension as need or not-needed.
- **Treat:** apply only the intervention matching the diagnosis.
- **Recover resources:** if blocked, suspend case, service at station, return, resume.
- **Report:** close only after full assessment and required treatment.

Sensing is separated from diagnosis. Treatments are diagnosis-gated, so generic watering/fertilizing is impossible in the safe

policy. **finish-assessment** forces complete classification before reporting.

Sample Problems

Plan	Behaviour
P1 single delayed-growth plot	One localized issue; healthy plots are inspected/reported without treatment.
P2 heterogeneous plots	Different plots receive water, pesticide, fertilizer/growth support, or no treatment according to local diagnosis.
P3 service recovery	Resource/battery shortage triggers suspend-service-return-resume-treat-report.

3. Q2: PDDL+

Numeric variables: moisture-level, pest-level, nutrient-level, growth-index, damage-level, toxicity-level, resource and battery fluents.

Processes: Continuous moisture decrease, pest spread, nutrient depletion, growth decline, damage accumulation, timed sensing, timed verification, and dynamic treatments: irrigation increases moisture, spray reduces pest pressure, fertilization restores nutrient level, growth support improves growth index.

Events: Threshold crossings create moisture-deficient, pest-infested, nutrient-deficient, growth-delayed, crop-failed, treatment-damaged, treatment target reached, resource depletion, and battery reserve violation.

In general, a discrete action starts sensing/treatment; a process changes numeric state over time; an event records target/threshold crossing; verification confirms recovery; reporting is enabled only when the necessary verified conditions hold.

- Delayed arrival allows continuous degradation and can make the original safe mission impossible.
- Pest, water, nutrient, and growth issues require different actions and resources.

- Unsafe/excessive irrigation/fertilization can trigger damage instead of improvement.
- Treatment start is not enough; the recovered state must be checked.
- Insufficient resource causes resource-limited reporting instead of unsafe treatment.

Sample Problems

Problem	Behaviour
P1 delayed failure	6-unit travel; crop fails during delay; robot observes and reports failure.
P2 wrong intervention	Robot diagnoses no treatment needed, then unsafe irrigation triggers damage.
P3 pest spread/spray	Pest evolves; pesticide is selected; spray process and verification precede report.
P4 nutrient/growth	Fertilize first, verify nutrients, support growth, verify growth, report.
P5 resource-limited	Water need detected but water below requirement; report resource limitation.

5. Conclusion and Limitations

The final model combines Q1 symbolic planning with Q2 hybrid PDDL+. Q1 demonstrates explicit inspection, diagnosis, targeted treatment, reporting, heterogeneous plots, and service recovery. Q2 demonstrates continuous condition evolution, threshold events, dynamic treatments, verification, delayed failure, wrong-treatment damage, recovery, and resource-aware decision-making.

Real agricultural systems include uncertain sensing, weather, soil variability, spatial pest diffusion, actuator error, and complex crop-growth dynamics. The PDDL model used is however deterministic and symbolic, while the PDDL+ model uses simplified linear rates and threshold events. Another limitation is planner scalability. The PDDL+ model can represent continuous crop dynamics and treatment effects, but adding full service recovery loops with refill and recharge caused severe search growth in ENHSP, suggesting a possible hierarchical planning.